



DSA in C++

Conditionals and Loops

Lecture 06

if(a) and if(a=b)

0 → false

1 → true

variable, assign, increment



if (condition) {

→ agar condition true
hav

}

else _____

↓ false

if(a) and if(a=b)

```
✓ int a = 0;  
✓ if(a=0) cout<<"Jitendra";  
else cout<<"Anand"; ✓
```

$\left[\begin{array}{c} 0 \\ \cancel{0} \end{array} \right]$
a

Output

if(a= something) → if(a)

if(a) and if(a=b)

```
✓ int x = 4;  
✓ if(x1++) cout<<x;  
else cout<<"Prem";
```

5

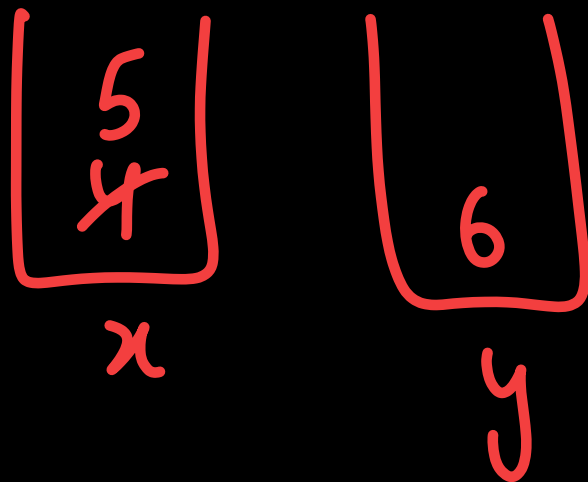
$\begin{pmatrix} 5 \\ 4 \end{pmatrix}$
x

if(a) and if(a=b)

```
✓ int x = 4, y = 6;  
  if(x++ || y++) cout<<x<<y;  
  else cout<<"Ayush";
```

Output

5 6



Ternary Operator

condition ? sach : jhoot

→ if-else but cooler

Ques: Take positive integer input and tell if it is odd or even

HW: Using nested ternary, print the greatest of three

Switch Case → alternate to if-else but (STUPID)

```
switch (number) {  
    case 1:  
        cout << "1";  
        break;  
    case 2:  
        cout << "2";  
        break;  
    :  
    :  
}
```


Ques: Given ith day of week , tell which day it is

1 → monday

2 → tuesday

3 .

4 .

5 .

6 .

7 → sunday

Loops

for, while, do while

Contents

1. For Loop
2. Questions
3. Break & Continue
4. While & Do-while Loop
5. Questions

What and Why?

loop
↓
repetition

For Loop

Q. Print "Good Morning" 10 times

Q. How many times will this loop run $[a \leq b]$

```
for (int i = a, i <= b, i++) {  
    cout << "Hello ? w" << endl;  
}
```

$b - a + 1$

Ques: Print _____ 'n' times. Take 'n' input from user

How **For** Loop works : the various parameters

initialization condition increment

```
for(int i=1; i<=3; i++){  
    cout<<"hello"<<endl;  
}
```

work/body

$\text{int } i = 1;$

$\boxed{1}$
 i

$i \leq 3$

$i++ = i = i + 1$

How For Loop works : the various parameters

```
for(int i=1; 4 <= 3; i++){  
    cout<<"hello"<<endl;  
}
```

4
~~3~~
~~2~~
~~1~~
i

✓initialisation

condition check

work

increment

repeat

fail

loop X

Output

hello

hello

hello

.

Ques: Print numbers from 1 to 100

Ques: Print all even numbers from 1 to 100

HW: Print all odd numbers divisible by 3
from 1 to 100

3 9 15 21 . . 99